

Ohm's law: $V = IR$

Resistance is directly proportional to voltage at constant temperature

Series and Parallel circuits

Series:

$$I_1 = I_2 = I_3 \quad (\text{current same})$$

$$V_{\text{total}} = V_1 + V_2 + V_3$$

$$R_{\text{total}} = R_1 + R_2 + R_3$$

Parallel:

$$V_1 = V_2 = V_3$$

$$I_{\text{total}} = I_1 + I_2 + I_3$$

$$\frac{1}{R_{\text{total}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

Basics of Electronics notes

✓ Electronics and charge

key formula: $Q = I \cdot t$

$Q \rightarrow$ charge (Coulombs, C)

$I \rightarrow$ current (Amperes, A)

$t \rightarrow$ time

- Conduction electrons - free electrons moving around atom cores in metals

- Conventional current - flows + $\rightarrow$ -
(opposite to electron flow)

✓ Potential Difference

Key formula: $V = \frac{W}{Q}$

$V \rightarrow$ voltage (Volts)

$W \rightarrow$ energy (Joules)

$Q \rightarrow$ charge (Coulombs)

Kirchhoff's First Law (KCL)

Statement: Sum of currents entering a junction = sum of currents leaving

Formula $\Rightarrow I = I_1 + I_2 + I_3$

Key points:

- Consequence of conservation of charge
- Current does not increase or decrease at junction
- Junction = point where ≥ 3 circuit paths meet.
- Branch $\rightarrow$ path connecting two junctions
- Series circuit $\rightarrow$ current same at every point
- Parallel circuit $\rightarrow$ current divides at junctions, each branch carries different value

Kirchhoff's Second Law (KVL)

Statement: Total EMF in a closed loop = sum of all potential differences across components.

Formula: $E_1 + E_2 = V_1 + V_2$

Key points:

- Consequences of conservation of energy
- Energy in = energy out
- Series circuit $\rightarrow$ total p.d splits across components proportional to resistance
- Parallel circuit $\rightarrow$ p.d same across each closed loop.
- Each closed loop acts as independent series circuit
- If one branch breaks off $\rightarrow$ current still flows through others

EMF and Internal Resistance

Key formulas:

$$E = V + Ir, \quad E = I(R + r),$$

$$\text{Lost volts} = Ir$$

Where:

$$E = \text{EMF}$$

$$V = \text{terminal p.d.}$$

$$I = \text{current}$$

$$R = \text{external resistance}$$

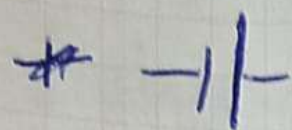
$$r = \text{internal resistance}$$

EMF vs P.D:

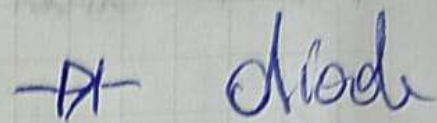
EMF $\rightarrow$ energy given to circuit by source

P.D $\rightarrow$ energy taken from by components

Some circuit symbols



Battery / cell



diode



Resistor



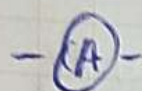
Potentiometer



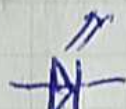
Switch



LDR



Ammeter



LED



Voltmeter

Potential Divider

Key formula = $V_{out} = V_{in} \cdot \frac{R_2}{R_1 + R_2}$

How it works:

- Two resistors in series across a supply
- Output voltage taken across R_2
- Changing R_1 or R_2 changes V_{out}

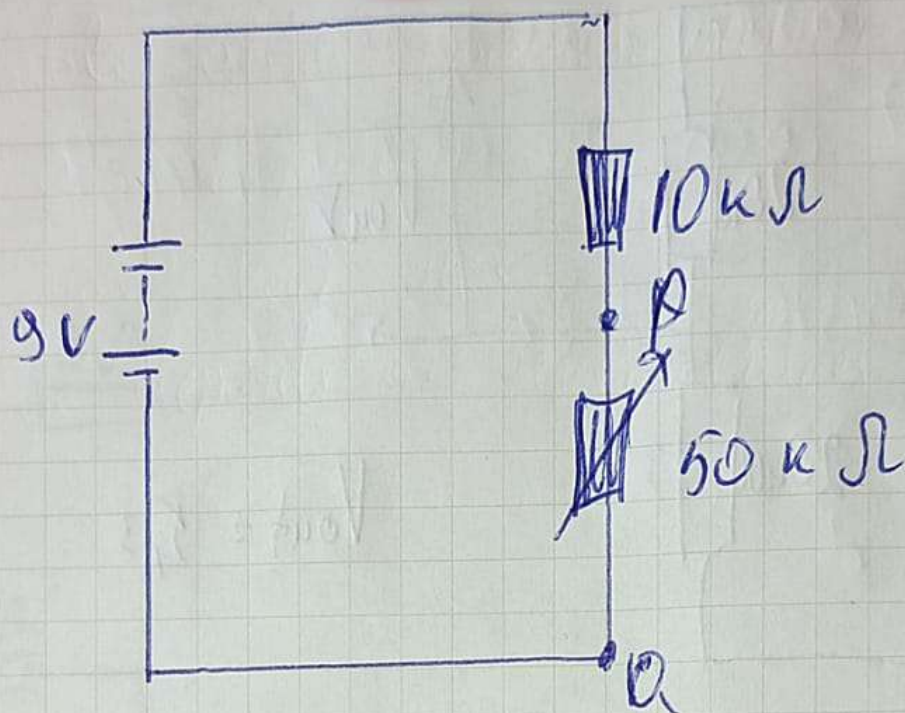
Variable resistor (potentiometer)

$$R_1 = 0 \rightarrow V_{out} = 0$$

$$R_1 = \text{max} \rightarrow V_{out} = \text{max}$$

Range depends on resistor values

3)



V_2 = potential difference between the terminals P and Q.

V_2 max:

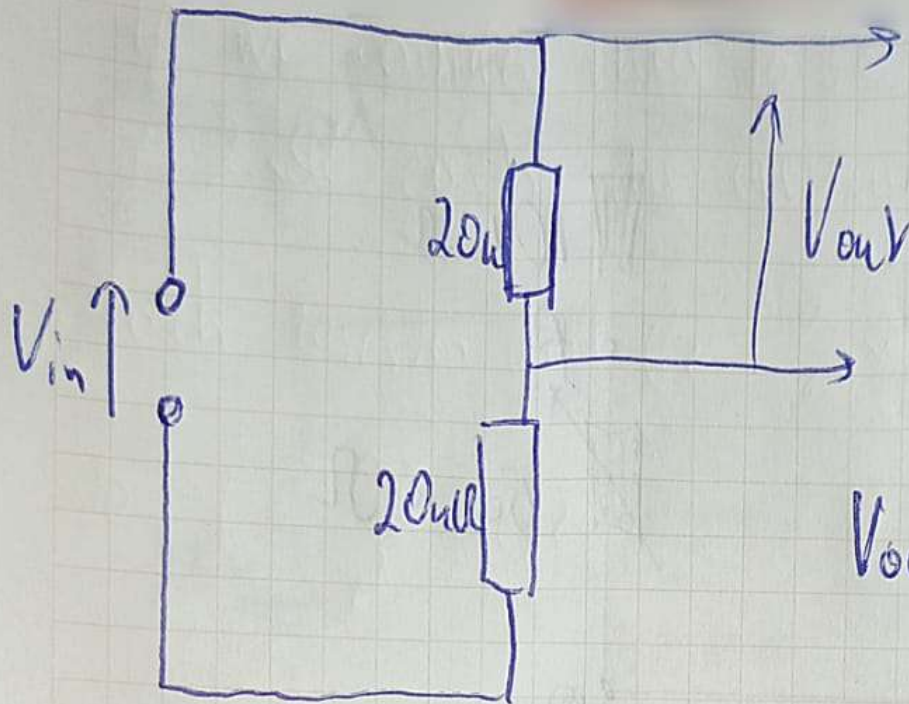
$$V_2 = 9 \cdot \frac{50}{50+10} = 7,5\text{ V}$$

min = 0V

V_2 min:

$$V_2 = 9 \cdot \frac{0}{0+10} = 0\text{ V}$$

max = 7,5V



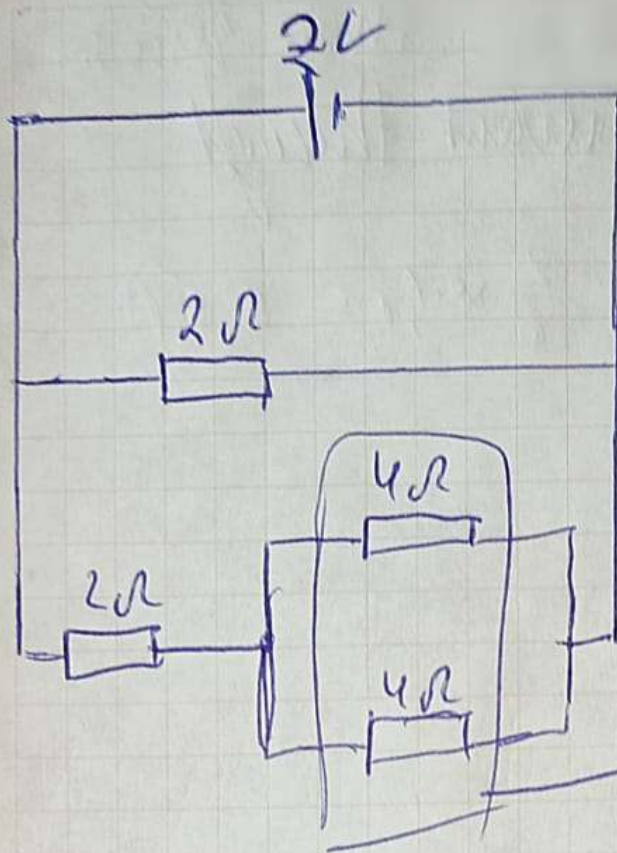
$$V_{out} = 5,3$$

$$V_{out} = V_{in} \cdot \frac{R_2}{R_1 + R_2}$$

$$V_{in} = \frac{R_1 + R_2}{R_2} \cdot V_{out}$$

$$V_{in} = \frac{R_1 + R_2}{R_2} \cdot V_{out} = \frac{20 + 20}{20} \cdot 5,3 = 10,6$$

$$\text{If } R_1 = 12k\Omega, R_2 = 10k\Omega \rightarrow V_{in} = \frac{12 + 10}{10} \cdot 5,3 = 8,5$$



$$\frac{1}{R_{t+1}} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{4} + \frac{1}{4} = \frac{2}{4} = \frac{1}{2}$$

$$R_{t+n} = 2\Omega$$

$$R_1 + R_{t+n} = 2 + 2 = 4\Omega$$

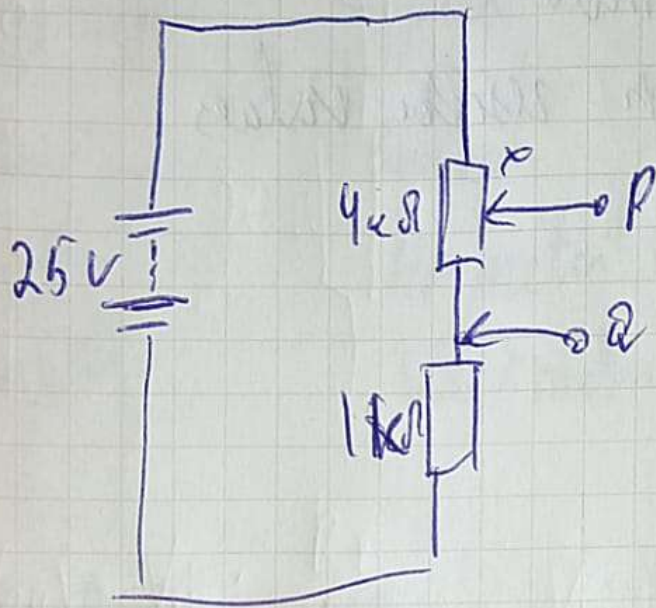
$$\frac{1}{R_{\text{total}}} = \frac{1}{R_{1+2+3}} + \frac{1}{R_n} = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$$

$$R_{\text{total}} = \frac{4}{3}\Omega$$

$$I = \frac{V}{R} = \frac{2V}{\frac{4}{3}} = 1,5A$$

Practical Questions

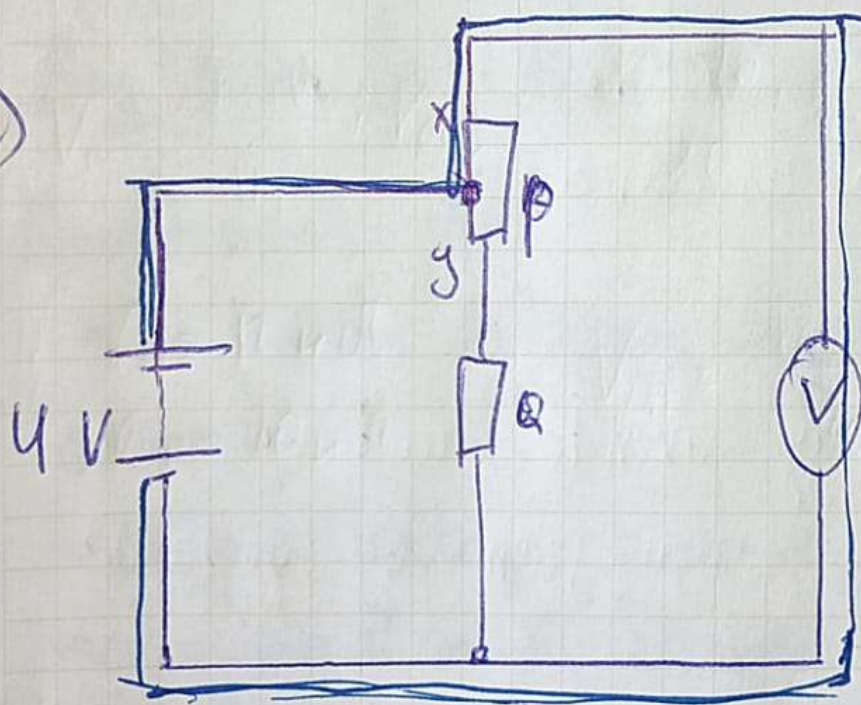
1)



$$V_2 = 25 \cdot \left(\frac{4}{1+4} \right) = 20V$$

$$V_2 = 25 \cdot \left(\frac{0}{1+0} \right) = 0V$$

2)



Voltmeter
always reads
4V.

Because because
currents never
flows on Q,

if we consider there to be three resistors
of $2\ \Omega$ each, the current through
each resistor will be $\frac{2}{6} \times 1.5 = 0.5\text{ A}$

so $I = \frac{0.5}{2} = 0.25$

Diode

A diode allows current to flow in only one direction. Acts like a one-way valve in a circuit - forward biased it conducts, reverse & biased it blocks.

Key formulas:

$V_d = 0.7V$ forward voltage drop

$V_L = V_s - V_d$ voltage across series

$I = \frac{V_L}{R}$ Current in the circuit

Reverse: $I \approx 0$ no current in reverse bias

Key concepts:

Three terminals: Base (B), Collector (C)
Emitter (E)

Acts as a switch: small I_i , controls
large I_o

Acts as an amplifier: output
signal is β times the input

NPN: current flows from Collector $\rightarrow$
Emitter when base is activated

Transistor

(NPN-BJT)

A transistor is a semiconductor device that can amplify or switch electronic signals. A small base current controls a much larger collector current.

Key formulas:

$$I_c = \beta \cdot I_b \quad \beta = \text{current gain}$$

$$I_e = I_c + I_b \quad \text{KCL at emitter node}$$

$$V_{be} \approx 0,6V \quad \text{turn on voltage (silicon)}$$

$$I_b = 0 \rightarrow \text{OFF} \quad \text{no base current} = \\ \text{transistor off.}$$

• Used in filters, timing circuits and power smoothing.

LDR - Light Dependent Resistor.

An LDR is a resistor whose resistance changes with light intensity. More light means lower resistance, allowing more current flow.

Key Formulas:

Light $\uparrow$ $R \downarrow$ - inverse relationship

$$V_{out} = V_{in} \cdot \frac{R_2}{(R_{LDR} + R_2)} \quad \text{- used in voltage divider circuit}$$

Dark: $R = M\Omega$ - very high resistance
Light: $R = 100\Omega$ - very low resistance

Key concepts:

LDR is a type of variable resistor - resistance depends on environment

Compared with a fixed resistor in a voltage divider to detect light levels

Applications: automatic street light alarm systems, camera sensors

Key formulas: concept.

Forward bias: anode (+) connected to positive terminal - current flows

Reverse bias: current is blocked - acts as open circuit

LED is a diode that emits light when forward biased

Zener diode maintains constant voltage - used as voltage regulator

Capacitors

A capacitor stores electric charge between two conducting plates. It releases this energy quickly when needed.

Formulas:

$$C = \frac{Q}{V} \quad E = \frac{1}{2} CV^2$$

$$C_{+} = C_1 + C_2$$

$$\frac{1}{C_{+}} = \frac{1}{C_1} + \frac{1}{C_2}$$

Key concepts:

Capacitance is measured in Farads (F)
usually μF or nF in practice

Capacitors pass AC but block DC

Parallel $\rightarrow$ total capacitance increases; Series $\rightarrow$ decreases

Electrical Power

Power is the rate at which electrical energy is transferred in a circuit. It tells us how much energy a component uses per second

Key formulas:

$$P = IV$$

$$P = I^2 R$$

$$P = \frac{V^2}{R}$$

$$E = Pt$$

Key concepts:

Unit of power is the Watt (W) - $1W = 1 \frac{J}{s}$

Higher resistance = more heat generated for the same current